
FINAL DELIVERY

DATA ENGINEERING · NLP · NYC

EMPIRE'S TASTE.

New York City gastronomic intelligence — real-time sentiment analysis of the NYC culinary scene combining press coverage, real diner reviews, and recipes.

APACHE AIRFLOW

PYSPARK

VADER NLP

NEXT.JS 15

FASTAPI

DOCKER

3 data sources · daily pipeline · 3 data layers

THE PROBLEM

New York's food scene generates **thousands of daily signals** — articles, reviews, recipes — but they are scattered, unprocessed, and disconnected.

- ▶ What dishes are trending *this week* in NYC?
- ▶ Are real diners satisfied with the restaurants the press praises?
- ▶ Which recipes are aligned with what the media is talking about?
- ▶ How does sentiment evolve week over week?

THE SOLUTION

Automated pipeline that collects, cleans, analyzes, and exposes NYC gastronomic intelligence — daily, no manual intervention.

3

DATA SOURCES

10+

KPIS COMPUTED

~10min

FULL PIPELINE

USER STORIES

Prioritized by dashboard impact · 3 actors · 7 stories

PRIORITY	ROLE	NEED
HIGH	Catering manager	Top-rated dishes by event type → competitive menu proposals
HIGH	Event planner	Filter dish suggestions by dietary restriction (vegan, gluten-free, halal)
HIGH	Restaurant manager	Sentiment trends by dish category — detect cuisines gaining or losing popularity
MED	Data analyst	Data quality metrics: missing fields, ingestion rate, record counts
MED	Event planner	Dishes mentioned alongside NYC neighborhoods → location-specific menus
LOW	Manager	Automated alerts on sudden sentiment drops → proactively update event menus
LOW	Data analyst	Compare recipe metadata vs. review sentiment → validate recommendation model

WHERE THE DATA COMES FROM

WEB SCRAPING

EATER NY

RSS/Atom feed from NYC's leading gastronomic media outlet.

- ✓ Up to 20 articles per day
- ✓ Title, summary, author, categories
- ✓ Canonical URL for traceability
- ✓ Publication date

WEB SCRAPING + GRAPHQL

OPENTABLE

Real diner reviews via OpenTable's internal GraphQL endpoint.

- ✓ Playwright (Firefox headless)
- ✓ Automatic session — no manual cookies
- ✓ Ratings: food, service, ambiance
- ✓ Free-text diner feedback

REST API

SPOONACULAR

Recipe API queried dynamically with the keywords the NLP identifies as trending.

- ✓ Free-text or ingredient search
- ✓ Nutritional and dietary metadata
- ✓ Health score, price per serving
- ✓ 6 API keys with automatic rotation

EATER NY

20 art/day · RSS Atom

+

OPENTABLE

~200 reviews · GraphQL



NLP → KEYWORDS

TF-IDF + spaCy vectors



SPOONACULAR

5 recipes/keyword

MEDALLION ARCHITECTURE

BRONZE

Raw data — exactly as received

Unprocessed JSON · date-partitioned (YYYY-MM-DD/) · basic ID deduplication · maximum source fidelity



SILVER

Clean data — ready for analysis

Parquet · HTML removed · stopwords filtered · NLP columns generated · types fixed · cross-deduplication



GOLD

Analytical data — ready for consumption

Parquet (Spark coalesce) · VADER sentiment · TF-IDF keywords · governance KPIs · 10+ storytelling aggregations

WHY 3 LAYERS?

Bronze preserves raw data forever. Silver makes it usable. Gold makes it dashboard-ready. If an analysis fails, Gold reruns from Silver — no re-download needed.

BRONZE FORMAT

JSON

SILVER/GOLD FORMAT

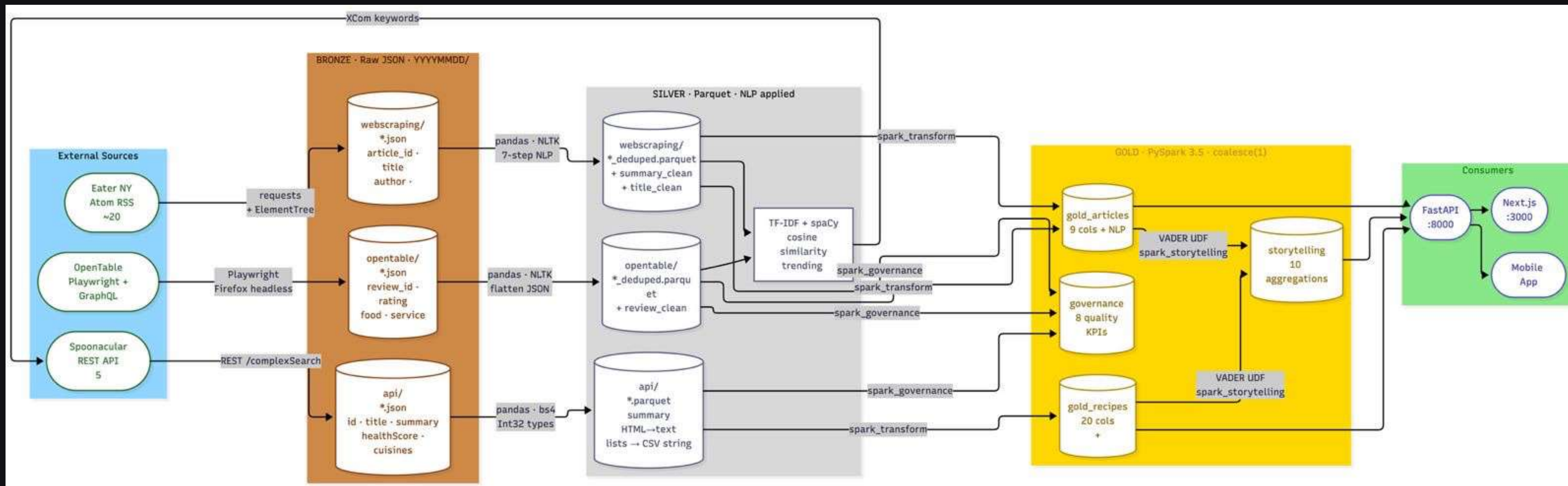
Parquet

GOLD ENGINE

PySpark 3.5

ORCHESTRATION

Airflow 2.10



THE PIPELINE — STEP BY STEP

DAG: `nyc_gastronomy_pipeline` · Manual trigger · Sequential · ~10 min end-to-end

#	AIRFLOW TASK	WHAT IT DOES	INPUT	OUTPUT
01	<code>extract_eater_ny</code>	Downloads articles from the Eater NY Atom feed	— (RSS feed)	<code>bronze/webscraping/</code>
02	<code>clean_eater_ny</code>	NLP pipeline: clean, dedup, stopwords → <code>article_summary_clean</code>	<code>bronze/webscraping/</code>	<code>silver/webscraping/</code>
03	<code>extract_opentable</code>	Playwright acquires session → paginates reviews per restaurant	— (web)	<code>bronze/opentable/</code>
04	<code>clean_opentable</code>	Flatten nested JSON → NLP → deduplicate by <code>review_id</code>	<code>bronze/opentable/</code>	<code>silver/opentable/</code>
05	<code>analyze_trends</code>	TF-IDF over articles + reviews → top 10 food keywords → XCom	<code>silver/webscraping/</code> + <code>opentable/</code>	<code>list[str]</code> keywords
06	<code>search_spoonacular</code>	Queries Spoonacular with each keyword received via XCom	keywords (XCom)	<code>bronze/api/</code>
07	<code>clean_recipes</code>	Flatten lists, strip HTML, cast types, dedup by recipe ID	<code>bronze/api/</code>	<code>silver/api/</code>
08	<code>prepare_gold</code>	Spark: transform + governance KPIs + storytelling aggregations	<code>silver/</code> (all layers)	<code>gold/</code> (4 Parquets)

SILVER LAYER — CLEANING & NLP

TEXT PIPELINE — ARTICLES

- | | | |
|---|-----------------------------|---|
| 1 | REMOVE PHOTO CREDIT | "Interior view. Photo: John
→" removed |
| 2 | DECODE HTML ENTITIES | ' → ' & → & |
| 3 | REMOVE URLS | https://... → [removed] |
| 4 | LOWERCASE + NO PUNCTUATION | "Best Ramen!" → "best ramen" |
| 5 | FILTER STOPWORDS | NLTK English + domain: "eater",
"nyc", "restaurant"... |
| 6 | REMOVE SHORT/NUMERIC TOKENS | 1-char tokens, purely numeric
→ removed |

Full example:

Raw: *"Outdoor dining. | Photo: Jane The best ramen spots in NYC opening this week"*

Clean: **"best ramen spots opening week"**

TRENDING KEYWORD EXTRACTION

CORE PROBLEM

From all tokens in an article, which ones are food and which are noise ("security", "terminal")?

HYBRID SOLUTION

- ▶ **TF-IDF** — rewards batch-distinctive tokens, penalizes ubiquitous ones
- ▶ **spaCy vectors** — cosine similarity vs. a 30-word food anchor
- ▶ **CUISINE_VOCAB** — curated fallback for "omakase", "birria", "vegan"...

REVIEW STOPWORDS (OPENTABLE)

Generic opinion words blocked from reaching Spoonacular:

amazing · wonderful · beautiful · favorite · staff · service · dinner · experience · recommend...

GOLD LAYER — ANALYSIS & INSIGHTS

VADER SENTIMENT ANALYSIS

VADER assigns a compound score (−1 to +1). Rule-based model designed for short media and social text.

$\geq +0.05 \rightarrow$ Positive	−0.05 to +0.05 → Neutral
$\leq -0.05 \rightarrow$ Negative	

Applied to Eater NY + OpenTable for balanced sentiment. Without OpenTable: always 100% positive — food press is inherently promotional.

KEYWORDS — TREND LABELS

TF-IDF SCORE	SENTIMENT	LABEL
≥ 80	positive	Highly trending
≥ 50	positive	Trending
≥ 30	positive	Growing
≥ 50	negative	Heavily criticized

10 AGGREGATIONS GENERATED IN GOLD

1	SENTIMENT_DISTRIBUTION	% positive/neutral/negative
2	SENTIMENT_TREND	Weekly mood evolution
3	TOP_KEYWORDS	Top 20 food keywords by TF-IDF
4	KEYWORD_SENTIMENT	Sentiment score per keyword
5	RECIPE_TREND_ALIGNMENT	Recipes aligned with trending topics
6	REVIEWS_SENTIMENT	Real diner sentiment (OpenTable)
7	REVIEWS_BY_RESTAURANT	Satisfaction ranking by restaurant
8	NAMED_ENTITIES	Restaurants, chefs, neighborhoods (NER)
9	DIETARY_BREAKDOWN	Recipes by dietary restriction
10	SOURCE_COMPARISON	Press vs recipes vs reviews

DATA GOVERNANCE

`spark_governance.py` — 8 quality dimensions over Silver data, persisted in Gold as long format (one row per KPI).

COMPLETENESS

% of null or empty values per column. Healthy threshold: < 20%

VOLUME

Total records and Parquet files generated on the day

UNIQUENESS

% of duplicates by primary key (recipe ID or article_id)

VALIDITY

% of expected columns present + IQR outlier rate

ACCURACY

NLP compression ratio + % non-ASCII characters (encoding noise proxy)

TIMELINESS

Number of daily partitions present in Bronze

OUTPUT FORMAT – LONG FORMAT

Single Parquet: `kpi_name` · `category` · `source` · `field` · `value` · `unit` · `finding`. Ready to query with any analytical tool.

Exposed in FastAPI as: `GET /api/governance/kpis` · `GET /api/governance/null-rates` · `GET /api/governance/outliers`

Empire's Taste — Data Governance Dashboard

Team: Data Engineering | Last updated: 2026-06-03 19:49:18

Total Records

43

Max Null Rate

97.0%

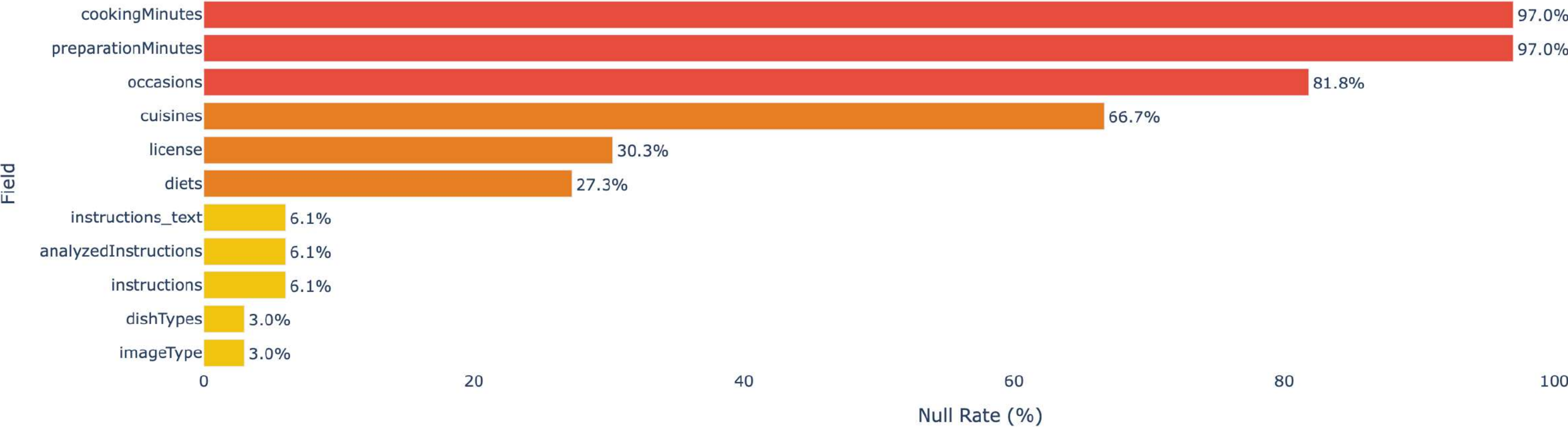
Duplicate Rate

0.00%

Schema Compliance

100.0%

Null Rate per Field — Silver Layer



Records Ingested per Day by Source



Outlier Rate per Numeric Field (IQR Method)



DEPLOYMENT — DOCKER

SERVICES IN DOCKER-COMPOSE

SERVICE	PORT	TECHNOLOGY	ROLE
airflow-scheduler	—	Airflow 2.10	Runs the DAGs
airflow-webserver	8080	Airflow 2.10	Monitoring UI
gastronomy-api	8000	FastAPI	Serves Gold data
web	3000	Next.js 15	User Dashboard
governance-dashboard	8050	Plotly Dash	Technical KPIs
storytelling-dashboard	8051	Plotly Dash	Trends
postgres	5432	PostgreSQL	Airflow Metadata

VOLUMES & CONFIGURATION

DATALAKE_BRONZE/	Mounted as volume → DAGs write here
DATALAKE_SILVER/	Shared between Airflow and API
DATALAKE_GOLD/	Shared between Airflow and API (FastAPI reads it)
AIRFLOW/DAGS/	Hot-reload: DAGs update without rebuild

FULL START

```
mkdir -p datalake_bronze datalake_silver datalake_gold
chmod -R 777 datalake_*
docker-compose up --build
```


THE WEB APP

/DASHBOARD/TRENDS

- ▶ Keyword of the day — live API: *"The city is eating bakery"*
- ▶ Sentiment **89%**) + mood label + score (**positive** interpretive phrase
- ▶ THE BOARD: keywords labeled Highly Trending / Trending / Growing
- ▶ Sentiment donut: press + diner reviews combined
- ▶ Weekly trend line + NER entity table (restaurants, chefs, neighborhoods)

/DASHBOARD/RECIPES

- ▶ Trending Spoonacular recipes selected by today's keywords
- ▶ Filters: cuisine, diet, prep time
- ▶ Cards with image, health score, time, ingredients
- ▶ Favorites system saved per user (Prisma + SQLite)

/DASHBOARD/FAVORITES

- ▶ Personal collection of saved recipes
- ▶ Persists per authenticated user (Google OAuth)

/DASHBOARD/DATA

- ▶ KPIs in user language: "Records collected", "Missing data", "Data quality"
- ▶ Pipeline layers explained in plain English
- ▶ Article counter with clickable titles → Eater NY
- ▶ Reviews by restaurant (OpenTable)
- ▶ Data completeness grid — field label + Healthy/Review needed badge

AUTHENTICATION

Google OAuth (NextAuth.js) — no passwords, dashboard-gated.

API — SERVICE LAYER

FASTAPI — LOCALHOST:8000

ENDPOINT	DESCRIPTION
GET <code>/api/storytelling/sentiment</code>	Combined distribution — articles + reviews
GET <code>/api/storytelling/keywords</code>	Top keywords with score and sentiment
GET <code>/api/storytelling/trend</code>	Weekly mood evolution
GET <code>/api/storytelling/entities</code>	Restaurants, chefs, neighborhoods (NER)
GET <code>/api/storytelling/sources</code>	Articles with links + reviews by restaurant
GET <code>/api/storytelling/summary</code>	Top keyword, % positive, total recipes
GET <code>/api/recommendations/</code>	Recipes filtered by cuisine/diet/time
GET <code>/api/governance/kpis</code>	Data quality KPIs
GET <code>/api/governance/null-rates</code>	Completeness by field

LOADER DESIGN

Reads Gold Parquet with `pandas.read_parquet()`.
`loader.py` auto-detects local filesystem (Docker) or S3 (AWS Lambda).

AUTOMATIC PARTITIONING

Always reads the latest `YYYY-MM-DD` partition. Falls back to yesterday if today's run is missing.

SENTIMENT FIX

Merges `sentiment_distribution` (Eater NY) + `reviews_sentiment_distribution` (OpenTable). Without this, result was always 100% positive.

Empire's Taste API

1.0.0

OAS 3.1

[/openapi.json](#)

NYC Gastronomic Trends & Sentiment — Gold layer REST interface

Governance

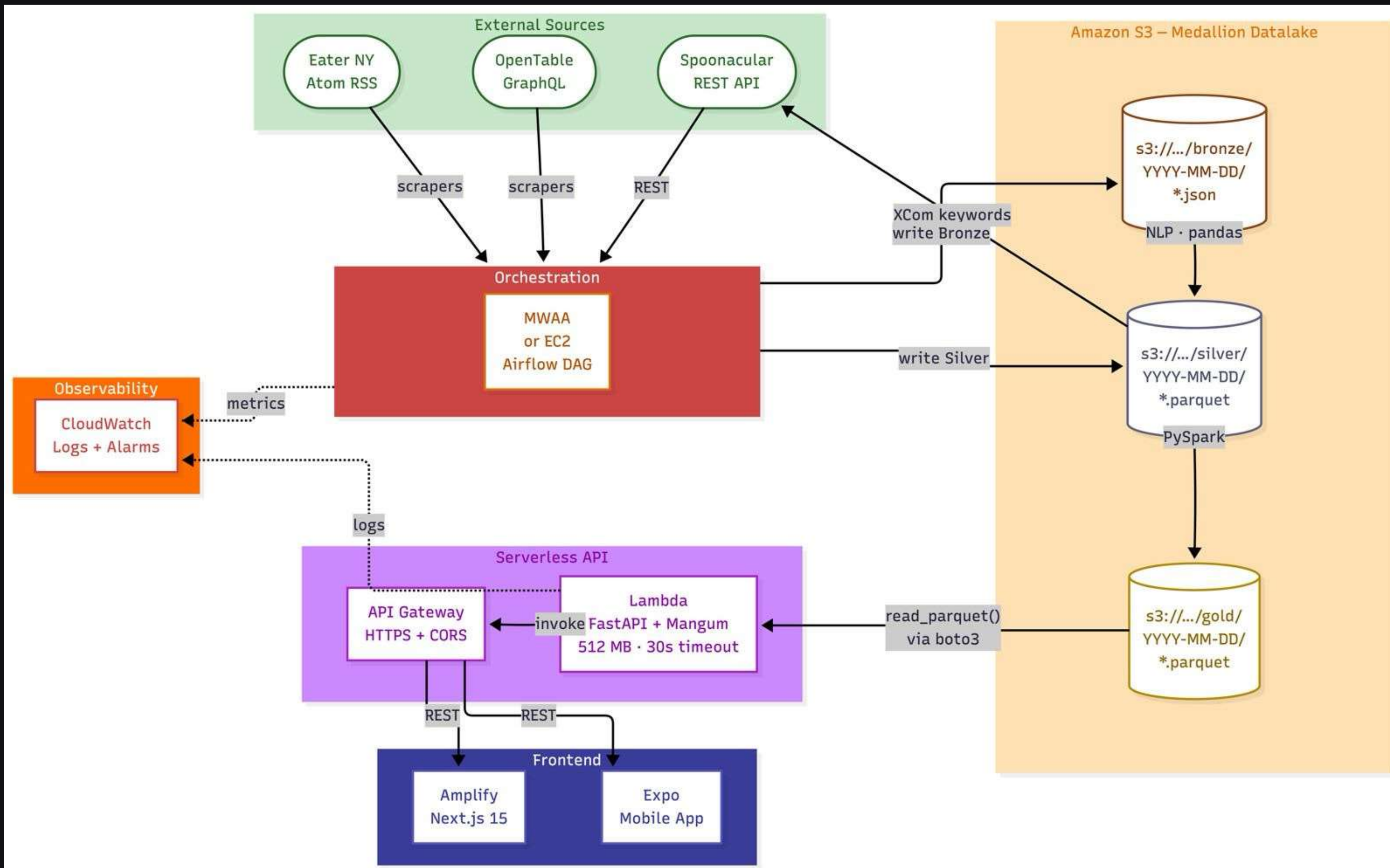


GET	/api/governance/	Get Governance	▼
GET	/api/governance/kpis	Get Kpi Summary	▼
GET	/api/governance/null-rates	Get Null Rates	▼

Storytelling



GET	/api/storytelling/	Get Storytelling	▼
GET	/api/storytelling/sentiment	Get Sentiment	▼
GET	/api/storytelling/keywords	Get Keywords	▼
GET	/api/storytelling/trend	Get Sentiment Trend	▼
GET	/api/storytelling/entities	Get Named Entities	▼
GET	/api/storytelling/summary	Get Narrative Summary	▼



EXECUTIVE SUMMARY

WHAT WE BUILT.

- ▶ End-to-end automated pipeline — Bronze → Silver → Gold — in Docker
- ▶ 3 sources unified: press (Eater NY) + reviews (OpenTable) + recipes (Spoonacular)
- ▶ NLP extracts real NYC food trends daily via TF-IDF + spaCy
- ▶ Balanced VADER sentiment: press + real diner reviews
- ▶ User dashboard in plain language + REST API + mobile app
- ▶ 8-dimension data governance automated with PySpark

3

DATA LAYERS

Bronze · Silver · Gold

8

DAG TASKS

Sequential, fault-tolerant

10+

GOLD AGGREGATIONS

Dashboard-ready

7

DOCKER SERVICES

One docker-compose up